

UEHR – USABILITY IN EHR

A Report submitted
in partial fulfilment for the Degree of
Bachelor of Technology
in
Computer Science and Engineering
by

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PROJECT WORK APPROVAL FOR B. TECH

The project work entitled ‘**UEHR-Usability in EHR**’ submitted by Gourav Bansal, Kishan Srivastava, and Vardan Agarwal to the National Institute of Technology, Delhi, is approved for the degree of Bachelor of Technology.

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CERTIFICATE

This is to certify that the project report entitled ‘**UEHR-Usability in EHR**’ submitted by Gourav Bansal, Kishan Srivastava, and Vardan Agarwal to the National Institute of Technology, Delhi, in partial fulfilment for the award of the Degree of **B. Tech in (Computer Science and Engineering)** is a *bona fide* record of project work carried out by him/her under my/our supervision. The contents of this report, in whole or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

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DECLARATION

I declare that this project report titled ‘**UEHR-Usability in EHR**’ submitted in partial fulfilment of the Degree of **B. Tech in (Computer Science and Engineering)** is a record of original work carried out by me under the supervision of **Dr. Shelly Sachdeva** and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice of reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

Electronic Health Records (EHRs) are digitalized representations of a patient's medical information, granting patients and healthcare professionals easier and quicker access to vital data when necessary. As EHRs gain widespread acceptance, the need for standardized EHRs becomes crucial, ensuring an effortless and universal exchange of health records while overcoming language barriers. Addressing this challenge necessitates achieving syntax, semantics, and language neutrality interoperability. For EHRs, Digital medical forms are essential, as they facilitate convenient information gain from a patient. But medical forms cannot be static; there can be the need to include some new fields in the form, and if the form is generated statically, adding new fields will require redesigning the whole form. So, it is essential to generate forms dynamically.

The main objectives of UEHR- Usability in EHRs:

- Generate medical forms dynamically from archetypes/templates.
- Provide language neutrality in the generated Form.
- Persist the form data in a suitable database.
- Terminology binding using SNOMED-CT.

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ABBREVIATIONS/ NOTATIONS

EHR	Electronic Health Record
CKM	Clinical Knowledge Manager
GUI	Graphical User Interface
ADL	Archetype Definition Language
XML	Extensible Markup Language
JSON	JavaScript Object Notation
UI	User Interface
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
API	Application Programming Interface
CEN ISO	European Committee for Standardization
FHIR	Fast Healthcare Interoperability Resources
HL7	Health Level 7
SNOMED CT	Systematized Nomenclature of Medicine Clinical Terms

CHAPTER 1

INTRODUCTION

1.1. EHRs and language barrier

Electronic Health Records (EHRs) are the digital version of a patient's health records, providing more accessible and faster access to information to the patient or a doctor in time of need.

EHRs are now becoming universal; thus, there is scope for the worldwide exchange of healthcare data, requiring syntax, semantic interoperability, and language neutrality.

For a country like India, where we have many regional languages, it is essential to build a system that is as easy for a person fluent in 'Hindi' language as for a person fluent in the 'Telugu' language while maintaining semantic interoperability.

Therefore, the user interface for the system should be available in various regional languages so everyone can use them comfortably.

1.2. EHR Standards

Over the year, many EHRs standards have been developed, such as OpenEHR[15], CEN(European Committee for Standardization) ISO(International Standardization Organization) 13606 [26], CDA (Clinical Document Architecture) by HL7(Health Level 7) [27], SNOMED CT[28](Systematized Nomenclature of Medicine Clinical Terms) and the latest development as FHIR (Fast Healthcare Interoperability Resources) by HL7[29].

These standards have adopted the two-level model, which separates the knowledge about the clinical data and the reference model into a two-level abstraction, achieving syntactic interoperability.

The reference model consists of general data structures and archetype models.

The OpenEHR and FHIR by HL7 are latest versions of EHR standards, and

Table 1.1 shows the information and the pros of the different developed EHR standards such as CEN ISO 13606, CDA by HL7 and SNOMED CT. Most of these standards are made to achieve semantic interoperability.

Our project embraces the openEHR standards as its foundation. Additionally, we leverage the power and benefits of SNOMED CT to enhance our system's functionality and semantic interoperability.

EHR Standards	About	Pros
1. openEHR	openEHR is an international standard and framework for designing and implementing health information systems. It aims to create a shared and standardized approach to capturing, storing, and exchanging electronic health records (EHRs) and other health-related data. The openEHR initiative was launched in 2000 with the goal of addressing the limitations of traditional EHR systems, which often suffer from proprietary formats, lack of interoperability, and limited flexibility.	• Ensures semantic interoperability • Open standard • Provides service-oriented interfaces • Flexible and adaptable framework for EHR systems
2. CEN ISO 13606	CEN ISO 13606 is a European standard that defines a communication framework for exchanging electronic health records (EHRs) between different healthcare systems. CEN ISO 13606 was developed by the European Committee for Standardization (CEN) and later adopted as an international standard by the International Organization for Standardization (ISO). It provides a common format and structure for exchanging clinical information, promoting interoperability and continuity of care.	• Ensures semantic interoperability • Focuses on capturing and exchanging clinical information accurately • Provides backward compatibility
3. CDA by HL7	CDA is a standard developed by Health Level Seven International (HL7) for the exchange of clinical documents. CDA provides a framework and structure for representing and sharing patient-centric information in a standardized and interoperable format.	• Widely adopted • Provides a structured document format that allows for the standardized representation of clinical data
4. SNOMED CT	SNOMED CT is a comprehensive clinical terminology and coding system that captures, represents, and exchanges clinical information across different healthcare domains and systems. SNOMED CT is recognised as one of the world's most extensive and widely adopted terminologies.	• Ensures consistent and unambiguous representation of clinical concepts • Provides extensive and comprehensive coverage of clinical concepts

Table 1.1 About Various EHR Standards and Their Pros

1.3. Semantic Interoperability

In the medical domain, thousands of medical terms are ambiguous, leading to different interpretations by different medical persons.

Semantic interoperability means the meaning of the data is the same for anyone who wants to use it.

Suppose a person has some medical records from a hospital and wants to get treated in another hospital. In that case, all his/her medical records must be transferred, and the other hospital should understand its semantics correctly.

A standardized approach for defining and structuring medical data is required to achieve interoperability, which can be achieved using archetypes.

1.4. Archetypes

Archetypes are standard models of a medical concept (e.g., blood pressure) used to achieve semantic interoperability. Archetypes are the maximal dataset of a medical concept, i.e., it contains everything related to the concept.

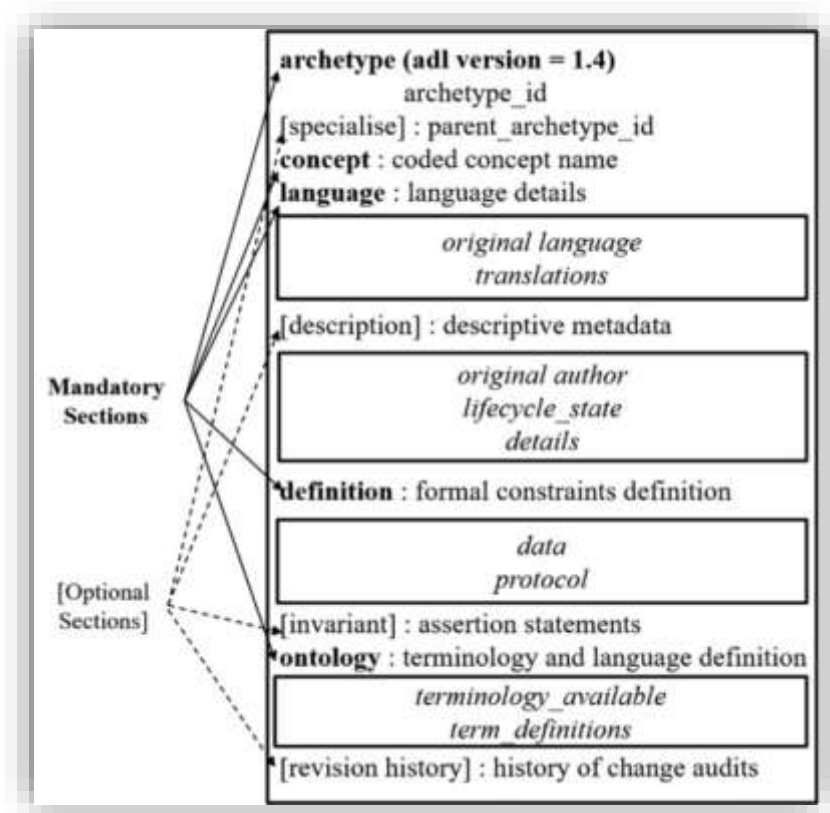


Fig. 1.1 Archetype Structure [1]

Figure 1.1 shows an archetype's top-level structure; the bold sections are mandatory, such as concept, language, definition, and ontology, whereas the sections enclosed with the square braces are optional, such as specialize, description, invariant, revision history, etc.

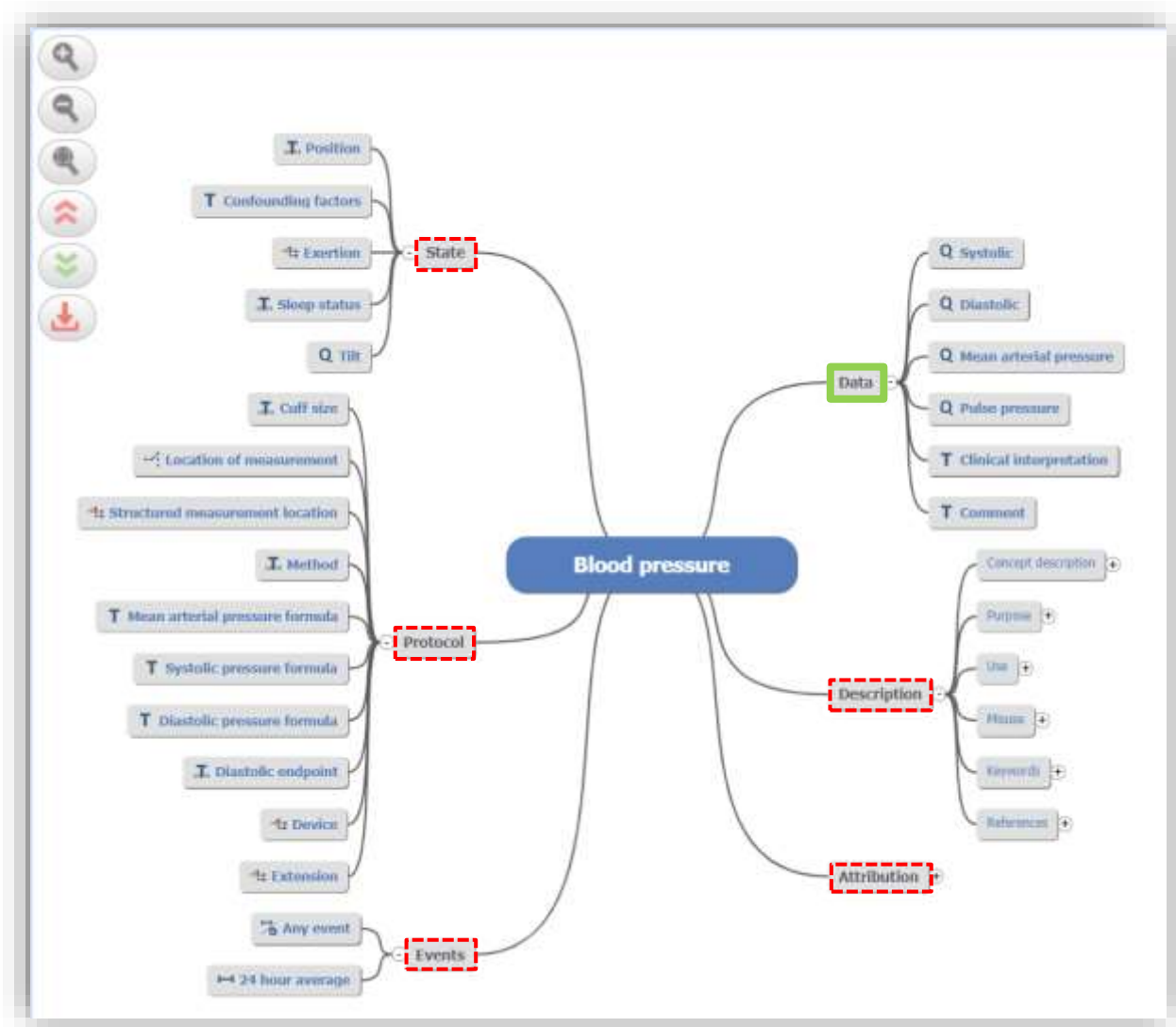


Fig. 1.2 Blood pressure as a concept [25]

Figure 1.2 presents a mind map illustrating the Blood Pressure concept. The mind map highlights the archetype's optional sections in red and dashed lines, while the mandatory section is depicted in solid green (data section). Following are the sections present in the concept –

1. **State**: Describes condition or status of the concept.
2. **Protocol**: Provides guidelines or procedures associated with the concept.
3. **Events**: Captures significant occurrences or incidents related to the concept.
4. **Data**: Includes attributes or fields defining the structure of the concept.

5. **Description:** Offers additional context and explanatory information about the concept.
6. **Attribution:** Identifies authors or contributors responsible for creating and maintaining the concept.

To provide clarity on the data types used by the parameters, symbols are placed alongside the parameters. For instance, the 'Q' symbol next to 'Systolic' indicates the Quantity data type (integer). While the 'T' symbol next to 'Clinical interpretations' represents the Text data type.

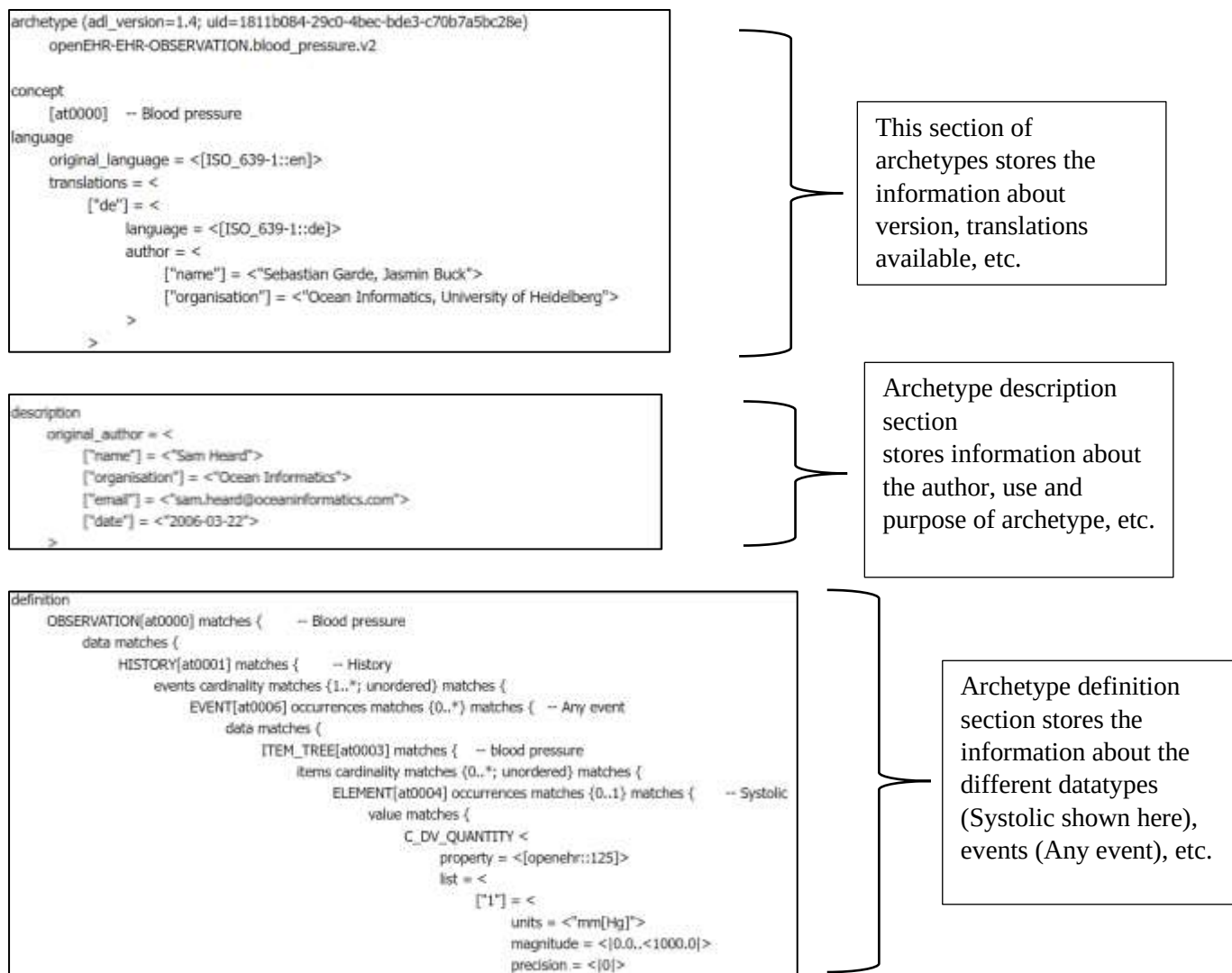


Fig. 1.3 Some of the sections of the Blood Pressure Archetype [24]

Fig.1.3 shows some sections of a 'blood pressure' archetype file. The first two images of Fig.1.3 shows the 'Attribution' and 'Description' sections, as mentioned in Fig.1.2. The Attribution section contains information related to the version, translations available, etc. The Description section contains information about the author, the purpose of the archetype, etc. The third image shows the 'Data' section, as mentioned in Fig.1.2 The Data section contains the information on attributes that define the archetype.

1.5. Usability

Healthcare applications should provide GUI in the language desired by the user.

Building GUIs for EHRs based on archetypes improves the healthcare applications' maintainability and reduces the developers' dependency.

Archetype models can help create medical forms that can capture the information required by doctors without relying on the developers.

Due to the language barrier, it is not easy in developing countries like India to take advantage of new healthcare technology in rural areas.

Thus, a user interface with multiple language support is highly in demand.

MOTIVATION

In recent years, the progress of payment systems in India has been remarkable, and a comparable rate of advancement is necessary for the digitalization of the healthcare industry. Since 2013, EHRs have been under development in India, and significant efforts have been made concerning Health Record IT Standards, Interoperability, and Data Ownership. However, many of these efforts lack usability. The recent rejection of the proposal to replace English with Hindi as the language for communicating with people who speak different languages underscores the need for multilingual implementation of all initiatives in a diverse country like India.

CHAPTER 2

STATE OF THE ART

Literature review

Papers	Summary	Pros	Cons/Challenges
Evolving Large Scale Healthcare Applications using Open Standards [1]	With standardization, the transition from one medical facility to another can be done instantly without any medical ambiguity or complexity.	• Eliminating language barriers. • Dynamic generation of forms. • Augmented databases to provide data storage in any language.	
Template4EHR: Building Dynamically GUIs for the Electronic Health Records Using Archetypes [2]	This paper discusses a tool to extract data attributes, terminologies, and constraints from an archetype to create a GUI using the extracted elements as input.	• Change in Archetypes updates the Form automatically. • Automatic creation of GUI's reduces dependency on Software Developers	• Does not discuss support for different languages
Semantic Interoperability in Standardized Electronic Health Record Databases [3]	Maintaining and standardizing EHRs using Archetypes-based databases to improve both data quality and semantic interoperability.	• Archetype-based model separates knowledge and information. • Archetypes are managed by clinicians, therefore, maintaining the quality semantics of data.	
Dynamic Generation of Archetype-Based User Interfaces for Queries on	The paper discusses the importance of a UI, how to generate a UI with archetypes, and the challenges faced while developing it.	• The proposed approach query language can perform most of the relational operations.	• The form generation is limited to only English language and doesn't provide solution for language neutrality.

Electronic Health Record Databases [4]	It also proposes a query language that can perform relational operations for querying		
Automated Creation of a Forms-based Database Query Interface [5]	The paper discusses the challenges of a form-based query system and how to tackle them. The biggest challenge is creating forms that can cover the maximum range of queries possible and be used by the users without being too complex and only with the help of database schema. The solution proposed in the paper introduces the queriability of entities, and attributes (which is the likeliness of an entity or attribute to be present in queries) and, based on queriability and the relationship between the entities or attributes, groups them together and creates a form that covers the considerable number of queries that users might need.	• The queriability parameter, mentioned in the paper, and its calculation can be used to create basic forms from archetypes, and medical experts can use and modify them and modify them as needed to generate other forms.	• The queriability parameter used to determine which attributes to include in the form, depends on the relations between different attributes. Sometimes, this relation may not be clear between the two or more attributes and thus queriability parameter will not be reliable in those situations.
Interactive browsing and Navigation in Relational Databases [6]	This paper proposes a presentation data model named ETable, which provides a presentation-level interactive view. This can help users construct join queries in relational databases. ETable can help users in interactive browsing and navigation of databases on an entity-	• Presents query result as an enriched table using Presentation Data Model. • It presents a Graph-Based Model which helps provide an abstraction of relational DBs. • Shows how users can	• It does not include complex query operations such as set operations.

	relationship level. It presents the result in an enriched table; each cell can have a set of entity references.	construct complex queries using Enriched Table. • Showcases a usable interface of ETable.	
Archetype-Based Electronic Health Records: A Literature Review and Evaluation of their applicability to health data interoperability and access [7]	This paper is based on the review of 47 papers and shows the development and implementation progress in archetypes.	• Real life use cases of archetype-based systems are presented. • Provides information about the challenges in developing EHRs and how archetypes can be solution to them.	
Distributed electronic health record based on semantic interoperability using fuzzy ontology: a survey [8]	This paper discusses the need for interoperability in EHRs, and some of the problems linked with it. It surveys the existing standards available to solve the problems arising in the path of interoperability. It also describes where these standards fail to achieve interoperability. It also discusses four different levels and categories of interoperability: No interoperability, syntactic interoperability, partial semantic interoperability and full semantic interoperability. EHR semantic interoperability requirements such as Health	• Discusses interoperability standards and challenges in semantic interoperability • Studies main interoperability problems • Surveys several results involved in achieving semantic interoperability using existing standards. • It suggests fuzzy ontology	• Doesn't discuss the application of fuzzy ontology. • Recommends technologies using fuzzy ontology such as FML and FDB but doesn't compare it with existing standards to prove the legibility of the suggestion of fuzzy ontology. Doesn't discuss the usability of fuzzy ontology in achieving semantic interoperability

	Domain Data types, mapping of ontology and costs are also discussed. It recommends fuzzy ontology as a solution. It discusses some capabilities of fuzzy ontology, like its ability to address real-world problems; in case of unclear data, it behaves like human thinking, and it is most suitable for the medical domain.		
OpenEHR modeling: Improving clinical records during the COVID-19 pandemic [9]	This paper discusses the implementation of a dual architecture that enables it to reuse data models developed in the OpenEHR community and tweak it as per the need for better results	• It shows the how the archetypes can be reused in different templates. • It shows the flexibility of openEHR where knowledge is changing constantly.	• Continuous development is needed for the effectiveness.
A semantic interoperability framework for distributed electronic health record based on fuzzy ontology [10]	This paper discusses an approach of semantic interoperability framework for distributed EHR base and handles the issues which occur during integration between heterogeneous systems.	• The framework proposed is flexible and can be adapted to different EHR systems. • The framework uses fuzzy ontology, which enables better semantic interoperability among distributed EHRs.	• Complexity is high.
Local EHR management based on openEHR and EN13606 [11]	The paper proposes a local electronic health record (EHR) management system based on openEHR and EN13606 standard which aims to provide	• Offers a unified model for EHR content and a platform for interoperability between different healthcare	• Lacks detailed discussion of the implementation and adoption challenges of the proposed system in real-world healthcare settings.

	a unified model for EHR content and a platform for interoperability between different healthcare systems. The system is designed to support the management and sharing of patient health data within local healthcare organizations, with the potentially improving the quality of care and reducing costs associated with redundant data entry and maintenance.	systems, enabling more efficient management and sharing of patient health data. • Has the potential to improve the quality of care and reduce costs associated with redundant data entry and maintenance. • Provides a practical solution for local healthcare organizations to manage and share patient health data more efficiently, using well-established standards and a common data model. • Offers insights into the benefits and limitations of openEHR and EN13606 standards for EHR management, providing guidance for future research and development in this area.	• Does not provide a comprehensive evaluation of the proposed system's functionality, usability, or effectiveness compared to other EHR management systems. • Relies on theoretical frameworks and case studies rather than empirical data, which may limit the generalizability of the findings to other healthcare settings.
Developing and Sustaining an Open Source Electronic Health Record: Evidence from a Field Study in Japan [12]	The paper presents a captivating case study of OpenMRS (Open Medical Record System), an open-source EHR system in Japan. It highlights the paramount significance of community involvement and	• Provides valuable insights into the development her sustainability of open-source electronic health record (EHR) systems in health care • Presents a case study of	• Relies heavily on qualitative data and may lack quantitative measures to support the findings. • Does not provide a comprehensive evaluation of the OpenMRS system's functionality, usability, or effectiveness compared to other EHR systems.

	collaboration in the development and sustainability of healthcare systems like OpenMRS.	OpenMRS, a successful open-source EHR system in a rural hospital in Japan, providing evidence of such systems' potential benefits. • Highlights the importance of community involvement and collaboration for the success of open-source EHR systems.	
Comparison of OpenEland HL7 FHIR Standards [13]	This paper explains and compares the two EHR standards, openEHR and HL7 FHIR on the basis of various parameters such as compatibility with systems, flexibility, etc.	• Provides a detailed explanation of openEHR and HL7 FHIR standards • Provides detailed comparative analysis of both the standards	
Performance of an openEHR based hospital information system [14]	This paper discusses the effect on the performance of healthcare system with openEHR-based software	• Compares openEHR server-based hospital information system with the traditional system	

Table 2.1 Literature Review

Research Gaps

1. **Syntactic Interoperability:** It is defined as the ability to exchange information.

This paper [3] proposes using standardized EHRs to achieve syntactic interoperability.

2. **Semantic Interoperability:** It ensures the meaning of terms remains the same for everyone. This paper [3] proposes using Archetypes to achieve semantic interoperability.

3. **Dynamic generation of GUIs:** This paper [2] proposes using the Template4EHR tool to generate forms using archetypes.

Technology Review

Tools/Technologies	Functionalities
Archetype Designer [16]	• It is a tool to create and export openEHR[15] Archetypes and Templates • Helps create templates by combining a set of Archetypes with different constraints as required. • Archetypes can be exported in different formats and can be uploaded to the Clinical Knowledge Manager Repository
Clinical Knowledge Manager [17]	• It is an online, available to all repository of clinical artefacts such as Archetypes, Templates, and Term sets. • Artefacts are constantly updated with the best clinical definitions enabling maximum interoperability. • Helps in managing and understanding clinical data models for different clinical content.
JSON [18]	• JSON (JavaScript Object Notation) is a data storage format. • JSON data is typically a name/value pair.
React Framework [19]	• React.js framework is a JavaScript framework for efficiently creating interactive UI and web applications.
JavaScript [20]	• JavaScript is a Programming language used mainly for web applications.
HTML [21]	• HTML (Hyper Text Markup Language) is used for creating web pages. • The browsers process this HTML file and render a UI for the user.
CSS [22]	• CSS (Cascading Style Sheets) is a design language used to style web pages.

Table 2.2 Tools Review

Research challenges

S. No.	Problems	Proposed Solutions/Implemented Solutions
1.	Syntactic Interoperability	Using standardized EHRs (openEHR Reference model)
2.	Semantic Interoperability	Using Archetypes and its terminology section
3.	Dynamic form generation from archetypes	We implemented the serialization of archetypes to JSON and identified fields to generate GUI.
4.	Language Neutrality	We provide language neutrality using available translations in an archetype
5.	Data persistence	We store the form data as JSON in a NoSQL-based databases.

Table 2.3 Research Challenges

CHAPTER 3

SOLUTION APPROACH

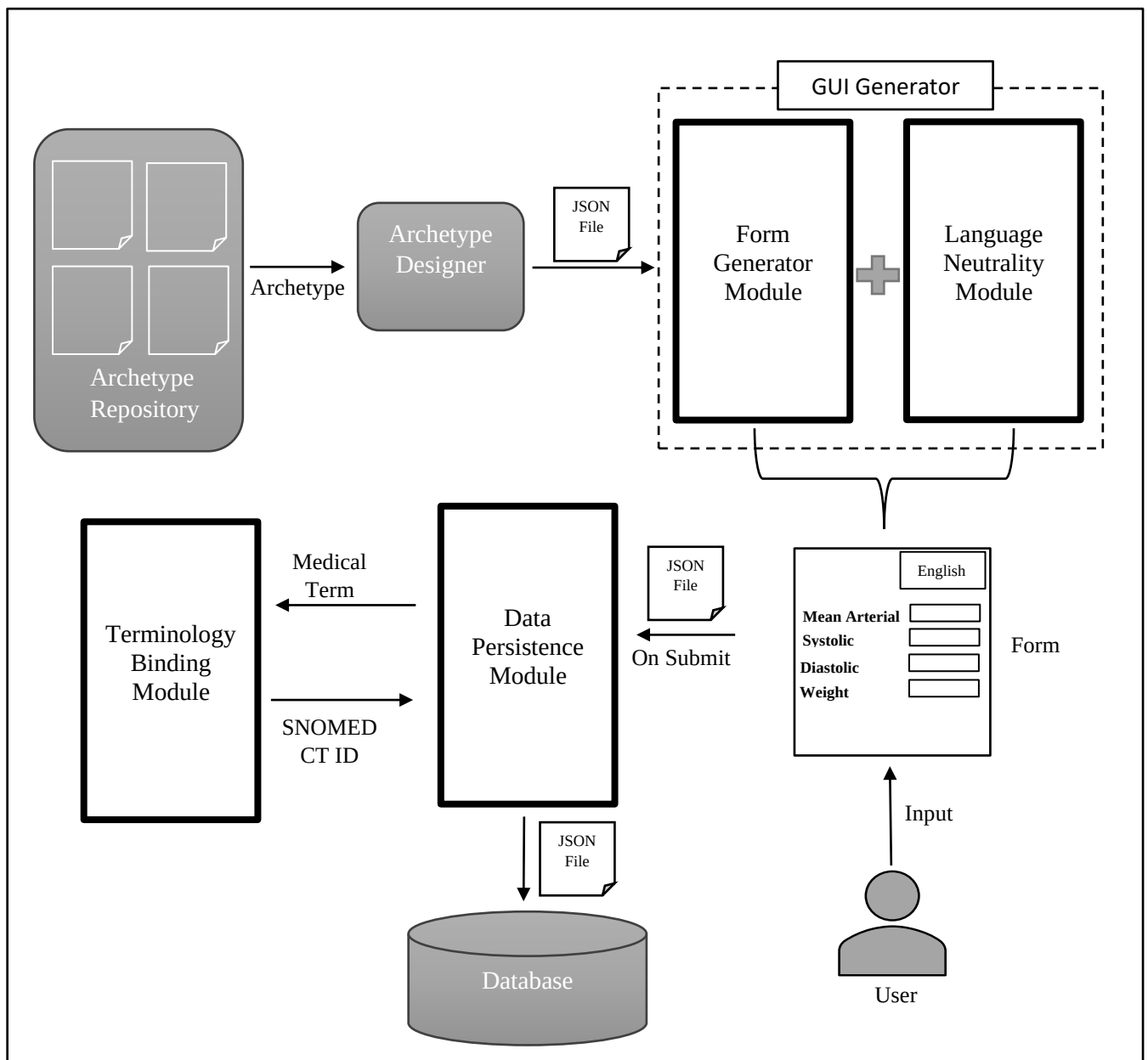


Fig. 3.1 Overall flow of the proposed solution

Fig 3.1 shows the overall flow of the proposed solution, which will be discussed in depth in this section. An ADL (Archetype Definition Language) compiler is used to convert it into Archetype Object Model (AOM) which is fed to a serializer which serializes to a JSON. This JSON acts as an input for GUI generation. GUI generator

takes JSON as input and generates a medical form. This form is language-neutral, and the language of the form can be changed after its generation. Fields not required in the form can also be removed after it is generated to reduce complexity. After submitting the form, Data Validation takes place, and concepts IDs are fetched using Snowstorm API corresponding to each medical term. Form data is stored in JSON along with the SNOMED CT concept IDs, and this is sent to be stored in MongoDB.

A detailed explanation of each module is explained in the following sections.

3.1. Generating medical forms dynamically from archetypes/templates

Various medical forms can be used, such as a form to store data regarding Blood pressure, Sugar levels, Pulse, etc. Each scenario requires a different archetype with different fields and structures, so it is difficult to build separate form for each archetype. Thus, a system is needed that can generate forms based on different archetypes dynamically.

3.1.1. Graphical User Interface Generation

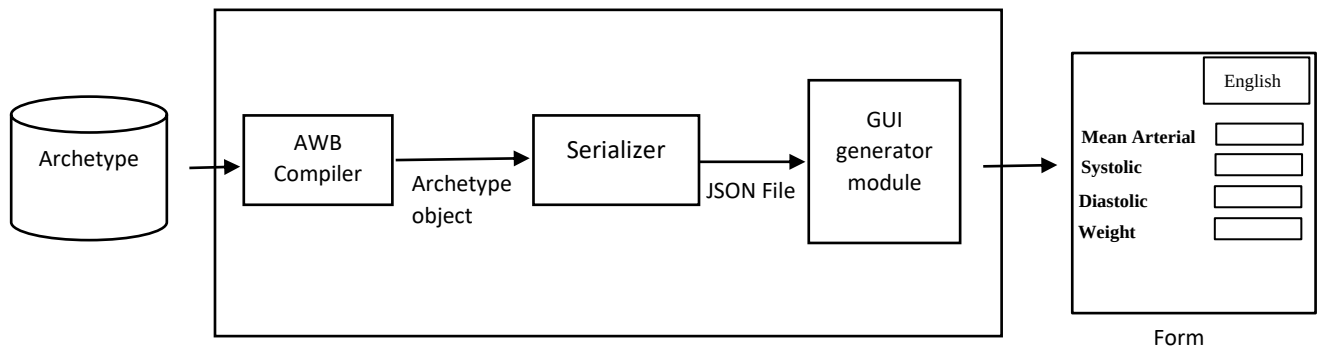


Fig. 3.2 Implementation Diagram of GUI Generation Module

We can generate a GUI with the help of an archetype. The GUI will be developed to support multiple languages supported by the archetype. To create a GUI, we convert the archetype into a template using the Archetype Designer Tool [16]. That template is passed to AWB [23] Compiler (ADL Workbench) as shown in Fig 3.2, which converts it into an Archetype Object Model (AOM), then serializes it into a JSON document. We get this JSON from the Archetype designer tool and pass it to our JavaScript program. It extracts all its data attributes, input constraints, and terminologies and uses them to generate a form GUI. Input Fields are defined for

every data type in the archetype. The data attributes are traversed, and input fields are generated based on the mentioned input constraints.

3.1.2. Mapping the datatypes to GUI Component

Mapping the datatypes to GUI Component means analyzing the input type for the data and allowing only the valid input formats.

E.g., if the temperature is the field to input the data, then only the decimal values should be allowed to enter.

The input type will be identified by looking for the ‘type’ tag of the information, and the UI will be generated correspondingly.

We have broadly classified three datatypes, DECIMAL, TEXT, and CODED TEXT. For the DECIMAL input type, only numerical values will be allowed; for the TEXT input type, textual input will be allowed; for the CODED TEXT, a dropdown representing the valid inputs will be shown for the user to select one.

3.2.1. Novelty of the project

UEHR (Usability in Electronic Health Records) introduces several novel aspects to enhance the usability of health records. Key novelty of the project is a combination of its three features: (1) UEHR incorporates a unique feature that generates medical forms in different languages. By leveraging archetypes, which are standardized clinical models, the system can dynamically generate forms tailored to specific languages. This capability helps healthcare providers cater to diverse patient populations and overcome language barriers, ultimately improving communication and patient care. (2) UEHR addresses the challenge of achieving semantic interoperability in health records. It achieves this by utilizing terminology binding with SNOMED CT (Systematized Nomenclature of Medicine Clinical Terms), a comprehensive clinical terminology system. SNOMED CT ensures that standardized, semantically rich terminology is used consistently across different healthcare systems and settings. This promotes accurate data exchange, effective data analysis, and enhances the overall interoperability of health records. (3) UEHR utilizes MongoDB, a NoSQL database, for data persistence. This choice of database technology provides scalability, flexibility, and high-performance data storage.

Overall, the UEHR project combines multilingual medical form generation, semantic interoperability through SNOMED CT, and data persistence using MongoDB to offer a comprehensive solution that improves the usability and effectiveness of electronic health records.

3.2. Providing language neutrality in the generated Form

Language Neutrality means supporting as many languages as possible to cover the majority of the population so that they can understand the form clearly even if they do not understand English.

3.2.1. Supporting languages provided in the ADL file

Most archetypes support multiple languages, such as English, Japanese, Spanish, French, Arabic, etc.

The translations are provided in most of the sections of an archetype such as descriptions, details, ontology, etc.

```
["en"] = <
  items = <
    ["at0000"] = <
      text = <"Blood pressure">
      description = <"The local measurement of arterial blood pressure which is a surrogate for arterial pressure in the systemic circulation,">
      comment = <"Most commonly, use of the term 'blood pressure' refers to measurement of brachial artery pressure in the upper arm,">
    >
  >
```

Fig. 3.3 The English translation of the ‘blood pressure’ term in Blood Pressure archetype

```
["ja"] = <
  items = <
    ["at0000"] = <
      text = <"血圧">
      description = <"全身を循環する動脈圧の指標として、局所で測定される血圧。一般的には「血圧」とは上腕で上腕動脈圧を測定したものをさすことが多い。">
    >
  >
```

Fig. 3.4 The Japanese translation of the ‘blood pressure’ term in Blood Pressure archetype

```

"aqlPath" : "/data[at0001]/events[at0006]",
"children" : [ {
  "id" : "systolic",
  "name" : "Systolic",
  "localizedName" : "Systolic",
  "rmType" : "DV_QUANTITY",
  "nodeId" : "at0004",
  "min" : 0,
  "max" : 1,
  "localizedNames" : {
    "ja" : "収縮期",
    "en" : "Systolic"
  },
},

```

Fig. 3.5 JSON file generated from the blood pressure archetype

Figure 3.6 shows the JSON file corresponding to the blood pressure archetype. For every term related to an concept, the archetype contains it's translated term as shown in Figure 3.5. Similarly, the corresponding JSON file also contains the translation under the 'localizedNames' section as highlighted in the figure 3.6.

Blood Pressure(OBSERVATION)

English ▼
Japanese
English

Patient ID

Patient Name

Any event

Systolic		mmHg ▼
Diastolic		mmHg ▼

Fig. 3.6 Sample form showing various languages to select from

Figure 3.7 shows a medical form for 'Blood Pressure' with language support for two languages for users to select. The languages in the JSON are tagged using the ISO 639-1 codes, and we are using the same ISO 639-1[31] codes to display the name of the language to the user.

ISO 639-1 is a standard published by the ISO (International Organization for Standards) that represents language names with a defined code. It provides a two-letter unique code for a language.

For example, ‘en’ corresponds to ‘English’, ‘ja’ corresponds to ‘Japanese’, etc.

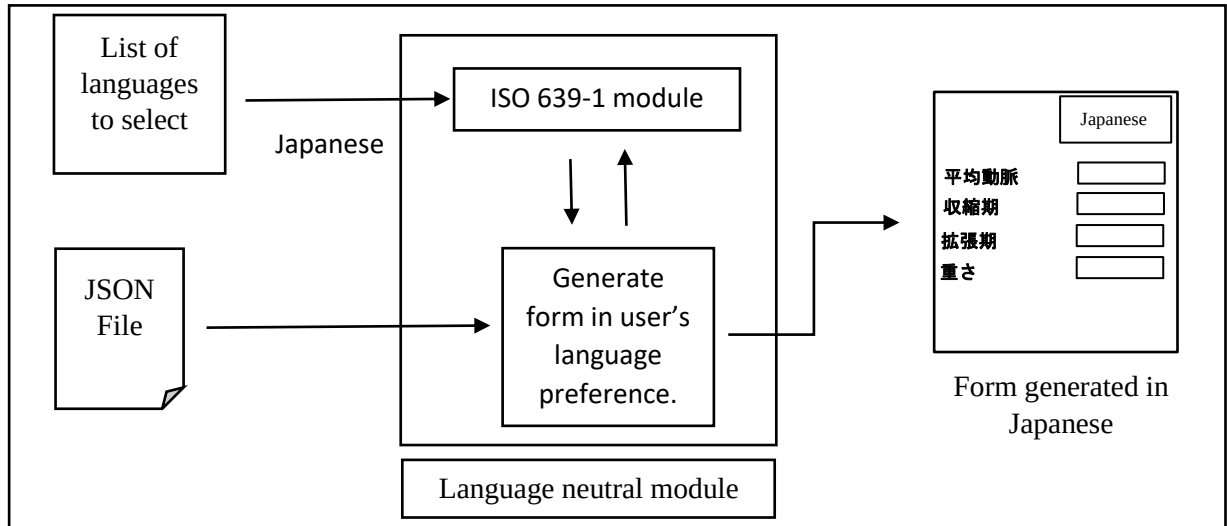


Fig.3.7 Component diagram of the Language Neutrality Module

Based on the user's language preference, the JavaScript program of the GUI generation phase picks up the translated version for each term from the ‘localized names’ section corresponding to the term from the JSON of the selected archetype and the GUI generation is carried out in the same manner as explained in sector 3.1.1. The process repeats whenever the user selects a different language from the dropdown, generating a Form in the new preferred language.

3.2.2. Adding new languages to the ADL file

Generally, an archetype is defined with support for multiple languages. Other original language translations can be added to the archetype to support localized languages.

It may be possible that the translation already exists in the CKM [17] repository. If it does not exist, the new language translation is defined by adding details to the translation section.

3.3. Persisting form data in a database

Data persistence means storing and retaining data beyond the lifetime of the process or application that creates it.

Data persistence can be achieved by storing data in a storage medium such as a hard disk, solid-state drives, or cloud storage. Techniques such as file-based storage, object-oriented databases, relational databases, or non-relational databases (NoSQL) are used to implement data persistence.

3.3.1. MongoDB database

MongoDB is an open-source document-oriented NoSQL database that uses JSON-like documents with schemas (optional) rather than traditional table-based relational databases [36].

MongoDB can store data as documents with different structures, which helps achieve flexibility and a scalable database system.

SQL (Relational database)	NoSQL (Non-relational database)
SQL databases are based on the relational data model and use tables to store the data.	On the other hand, NoSQL databases can store data in various data models, such as documents, key-value, column-family, or graphs.
They follow a proper schema, which is followed by the data.	NoSQL databases are generally schema-less, i.e., they can store data with different structures in the same collection/document.
Relational databases are horizontally scalable.	Non-relational databases are vertically scalable.

Table 3.1 Table of comparison between SQL and NoSQL

Other NoSQL databases include Google Firebase, Cassandra, Redis, and others. The reason to choose MongoDB is:

- MongoDB provides native support for JSON files, so it is much easier to store the health record data as JSON.
- The MongoDB database can easily be scaled vertically to handle large volumes of health records.

- The archetypes are prone to changes since medical knowledge evolves; hence the flexible schema of MongoDB ensures that health records can be stored regardless of any changes in the archetype.
- The flexibility of schema also ensures that any archetype (Observation, composition, instruction, etc.) can be used to generate Form.

3.3.2. Component diagram

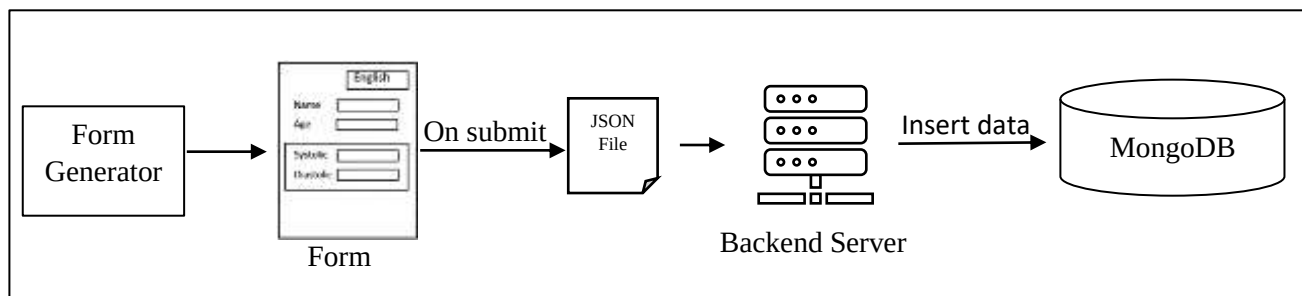


Fig. 3.8 Component diagram of the Data Persistence module

Fig3.9 shows the component diagram of the Data Persistence Module. The form generator program mentioned previously contains a ReactJS function that converts the form data into a JSON and makes a POST request on our backend server program whenever the user submits the Form using the submit button.

The form data then goes as a JSON file to our backend server, written in Node.js runtime environment using Express JS framework and stored in a MongoDB database.

3.4. Terminology Binding

The medical terms in the form, e.g., 'Systolic Blood Pressure', can be represented in various languages. Hence, there is a need to provide interoperability between the medical terms in various languages to avoid confusion while persisting data i.e., to achieve semantic interoperability, we have used SNOMED-CT terminology binding.

3.4.1. Introduction to SNOMED CT

SNOMED CT (Systematized Nomenclature of Medicine - Clinical Terms) is a comprehensive clinical terminology system representing clinical information in

electronic health records (EHRs). It is one of the world's most extensive, multilingual, and computer-processable medical vocabularies.

SNOMED CT is a structured and standardized system that includes over 350,000 active concepts and over 1 million unique descriptions of clinical information. It enables interoperability between different healthcare systems and facilitates the accurate and consistent clinical data exchange. SNOMED CT is used in various healthcare settings, including hospitals, clinics, research organizations, and public health agencies.

SNOMED CT has replaced many other terminologies, such as ICD-10 (International Classification of Diseases, 10th Revision), CPT (Current Procedural Terminology), and LOINC (Logical Observation Identifiers Names and Codes). It is recognized as a critical electronic health record infrastructure component, and many governments worldwide have endorsed its adoption.

3.4.2. SNOMED CT integration

SNOMED CT supplies two APIs for the terminology services: Snowstorm API [37] and the FHIR API [33]. Since our project is based on OpenEHR standards, we have used the Snowstorm server (collection of various endpoints) for our terminology binding requirements. SNOMED CT provides the facility to host the Snowstorm server locally over a user's machine, so we have used a locally hosted Snowstorm server to fulfil terminology binding.

Our project relies on the endpoint `"/snomed/{branch}/concepts"` to retrieve relevant data from the Snowstorm server.

```

{
  "items": [
    {
      "conceptId": "64882008",
      "active": false,
      "definitionStatus": "PRIMITIVE",
      "moduleId": "900000000000207008",
      "effectiveTime": "20020131",
      "fsn": {
        "term": "Increased body temperature (finding)",
        "lang": "en"
      },
      "pt": {
        "term": "Increased body temperature",
        "lang": "en"
      },
      "id": "64882008",
      "idAndFsnTerm": "64882008 | Increased body temperature (finding) |"
    }
  ],
  "total": 809,
  "limit": 1,
  "offset": 0,
  "searchAfter": "WzY0ODgyMDA4XQ==",
  "searchAfterArray": [
    64882008
  ]
}

```

Fig. 3.9 A sample response from the Snowstorm server

The Snowstorm server and MongoDB database allow for efficient and secure medical data management, making it easier to process and utilize data for various purposes. This approach also provides greater flexibility, as it is not tied to any software or platform and can be integrated with existing systems as required. Overall, this approach offers a scalable and effective solution for managing medical data and can be customized to meet the needs of individual healthcare organizations and applications.

3.4.3. Component Diagram

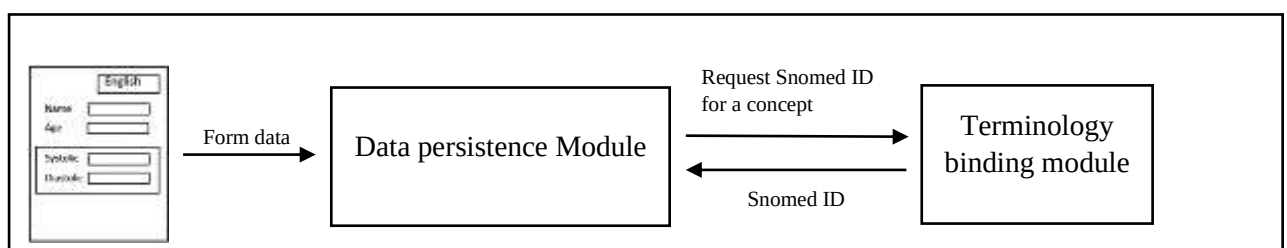


Fig. 3.10 Component diagram of the Terminology Binding module

The data persistence module consists of a ReactJS program, which requests the Snowstorm server to get SNOMED CT term IDs and makes a request to the backend server that puts the data into the database. The backend server and database are also part of the same module. The terminology binding module consists of the Snowstorm

server and a ReactJS program that takes the concept ID request and fetches the Snomed CT ID from the Snowstorm server.

3.4.4. Implementation Diagram of the Terminology Binding Module

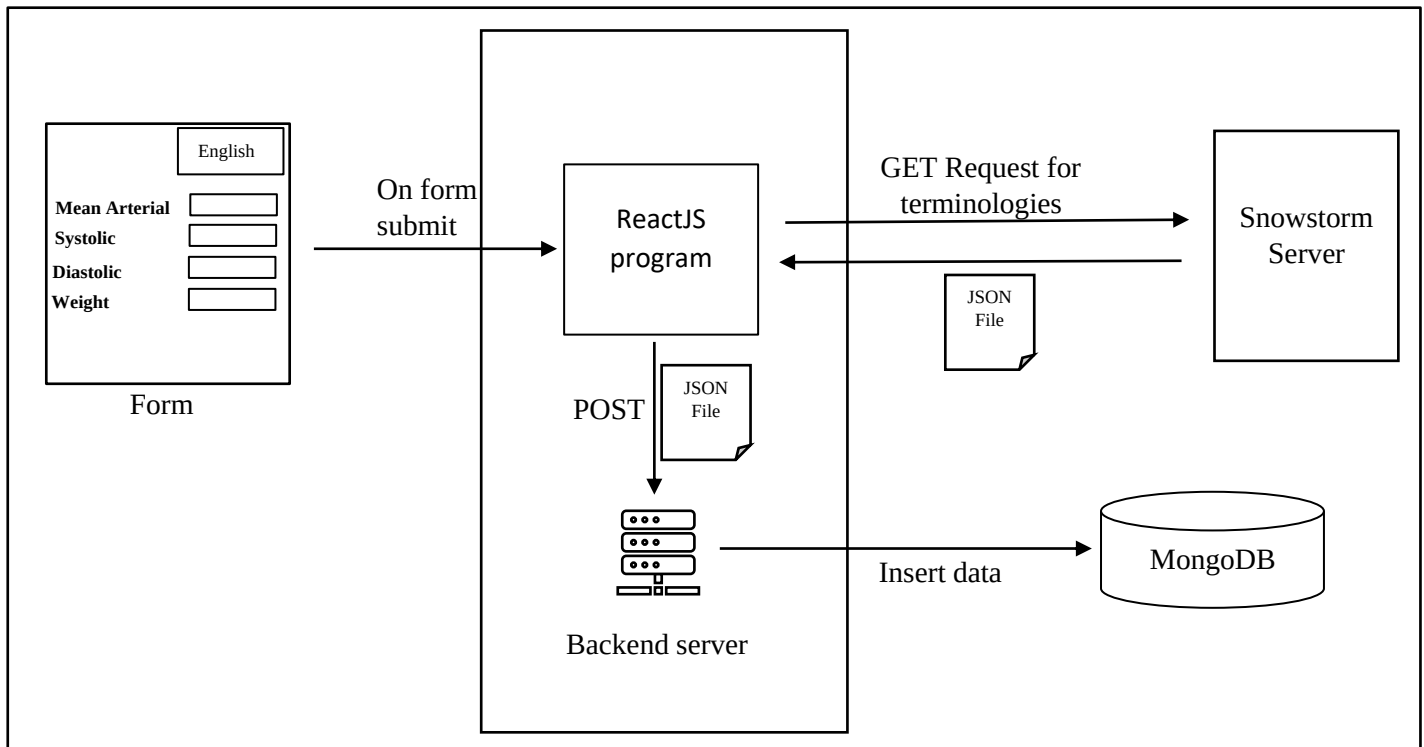


Fig. 3.11 Implementation diagram of the Terminology Binding module

When user submits the form, the form is converted as a JSON and goes to the data persistence module. Then a GET request is made to the Snowstorm server for each of the medical terms in the form, which returns a JSON response. This JSON contains concept id corresponding to the medical term. The concept id is inserted in the form JSON that is to be persisted in the database. Then the JSON is saved in MongoDB, which can be accessed and used as required.

Sample Request URL:

<http://localhost/snomed/MAIN/concepts?term=fever&offset=0&limit=1>, which shows a GET request URL for the terminology 'fever'.

CHAPTER 4

RESULTS/IMPLEMENTATION

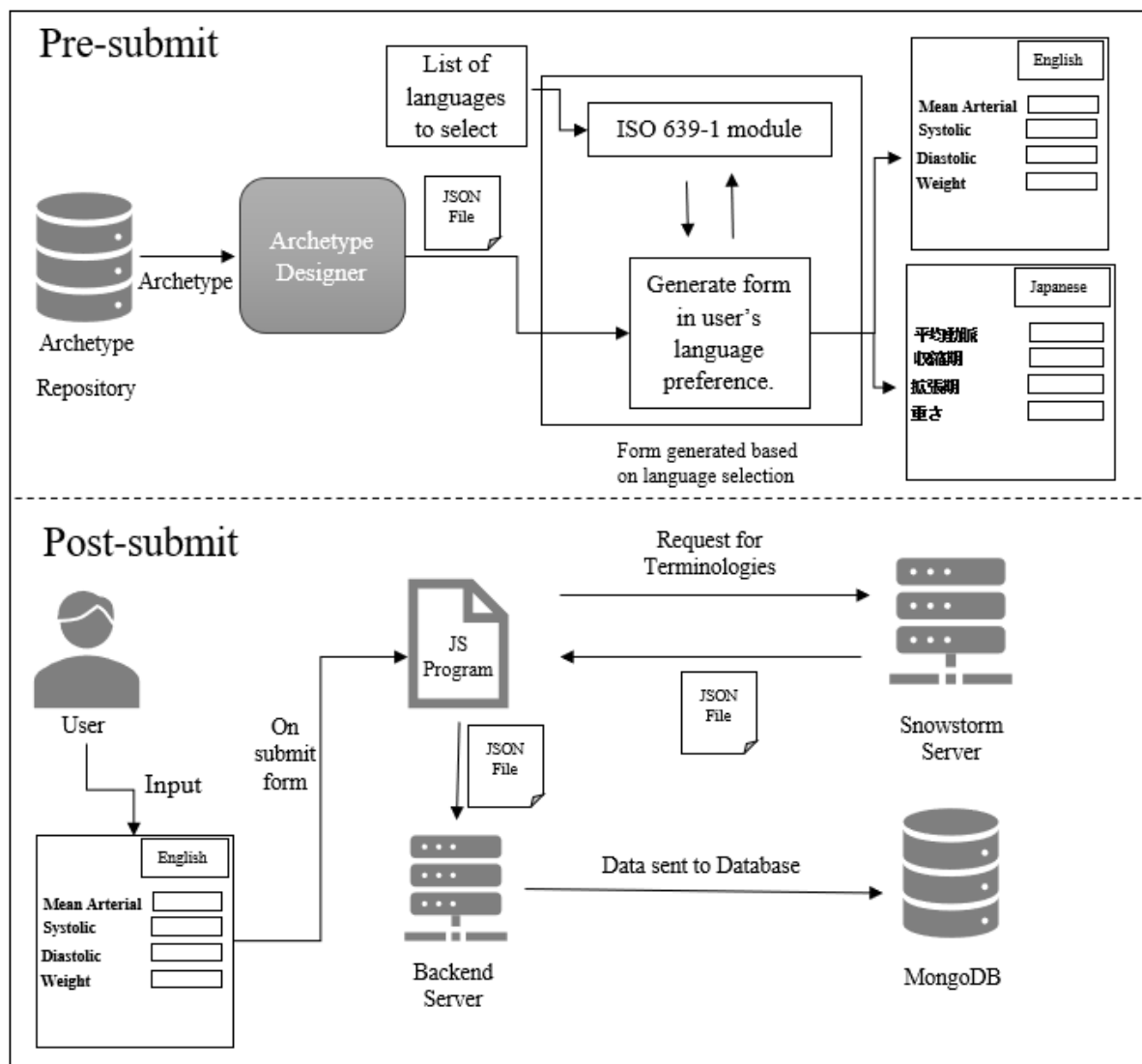


Fig. 4.1 Implementation diagram of the proposed system

Fig 4.1 shows the complete implementation diagram of the proposed system. We have divided the implementation into two parts, pre-submit and post-submit. The pre-submit part uses Dynamic Form Generation and the Language Neutrality modules, and the post-submit part uses the Data Persistence and Terminology Binding module. In the pre-submit phase, the form is generated dynamically, and the user can select a preferred language from the set of available languages to view his/her form in that language. Then after the user submits the form, the post-

submit phase starts, the medical terms from the form are linked to a SNOMED CT ID, and finally, the form data is persisted in a database.

The functioning of each module is explained in detail in Chapter 3 of the report.

This section further shows the results achieved with the proposed system. However, knowing our system's minimum hardware and software requirements is essential first.

4.1. Hardware and software requirements for the system

The minimum system requirements to use our system are as follows:

4.1.1. Hardware Requirements

- 8 Gigabytes of RAM
- 5 Gigabytes of ROM

4.1.2. Software Requirements

- Windows 10 or above, MacOS 11.1 or later
- Java 11 or more [38]
- Node.js 16+ [39]
- Snowstorm Server [40]
- React.js [19]
- Express.js [41]
- Elasticsearch 7.1.0 [42]
- MongoDB Community Server [43]
- Nginx [44]

4.2. Dynamic form generation and language neutrality

Language neutrality at a global level is achieved using the translations available in the archetypes however, at a regional level it is yet to be achieved in the project.

```
archetype (adl_version=1.4; uid=b5c14ff7-930d-4ae7-85c1-85c90747883e)
  openEHR-EHR-OBSERVATION.blood_pressure-bloodPressure.v2
  specialize
    openEHR-EHR-OBSERVATION.blood_pressure.v2

  concept
    [at0000.1]

  language
    original_language = <[ISO_639-1::en]>
    translations = <
      ["ja"] = <
        language = <[ISO_639-1::ja]>
        author = <
          ["name"] = <"Shinji Kobayashi">
          ["organisation"] = <"Kyoto University">
          ["email"] = <"skoba@moss.gr.jp">
        >
      >
    >
  >
```

Fig. 4.2 Blood Pressure ‘adl’ File

Fig. 4.2 shows the Blood Pressure ‘adl’ file snippet, which supports two languages: English and Japanese.

The ‘adl’ file is fetched from the clinical knowledge manager.

```

"aqlPath" : "",
"children" : [ {
  "id" : "any_event",
  "name" : "Any event",
  "localizedName" : "Any event",
  "rmType" : "EVENT",
  "nodeId" : "at0006",
  "min" : 0,
  "max" : -1,
  "localizedNames" : {
    "ja" : "任意のイベント",
    "en" : "Any event"
  },
  "localizedDescriptions" : {
    "ja" : "デフォルトで、テンプレートや実行時に明示的に定義されていない任意の時点や区間を示す。",
    "en" : "Default, unspecified point in time or interval event which may be explicitly defined in a template or at run-time."
  },
  "aqlPath" : "/data[at0001]/events[at0006]",
  "children" : [ {
    "id" : "systolic",
    "name" : "Systolic",
    "localizedName" : "Systolic",
    "rmType" : "DV_QUANTITY",
    "nodeId" : "at0004",
    "min" : 0,
    "max" : 1,
    "localizedNames" : {
      "ja" : "収縮期",
      "en" : "Systolic"
    },
    "localizedDescriptions" : {
      "ja" : "全身の動脈血圧での最高値 - 心臓の収縮期で測定される",
      "en" : "Peak systemic arterial blood pressure - measured in systolic or contraction phase of the heart cycle."
    }
  }
]
}
]

```

Fig. 4.3 Generated JSON File

Fig. 4.3 shows the JSON file snippet generated from the Blood Pressure ‘adl’ file. The JSON file contains the information in two languages: English and Japanese. For example, In Fig. 4.3, Under the ‘localizedNames’ section, ‘Systolic’ is written in two languages, English (en) and Japanese (ja).

The figure consists of two screenshots of a web form. The top screenshot shows the 'Resource Type' dropdown set to 'Observation' and the 'Archetype File' dropdown set to 'Blood Pressure'. The 'Archetype File' dropdown is open, showing a list of options: 'Blood Pressure', 'ECG', 'Pulse Oximetry', 'Respiration', and 'Spirometry'. The bottom screenshot shows the same form with the 'Archetype File' dropdown set to 'Blood Pressure' and the 'Submit' button visible.

Fig. 4.4 Selecting the Archetype to generate the Form

Fig. 4.4 shows the GUI where the user can select the resource type and the archetype file from the available archetypes, corresponding to the type of archetype selected, and its corresponding Form will be generated.

So, if the user selects Observation resource type and Blood Pressure archetype then the following Form will be generated:

The screenshot shows a web form titled "Blood Pressure(OBSERVATION)" with a language dropdown set to "English". Below the title, there are two input fields: "Patient ID" with the value "4519" and "Patient Name" with the value "Levi". A section titled "Any event" contains four rows of input fields. Each row has a label (Systolic, Diastolic, Mean arterial pressure, Pulse pressure) and a unit dropdown menu set to "mmHg".

Blood Pressure(OBSERVATION)	
Patient ID	4519
Patient Name	Levi
Any event	
Systolic	mmHg
Diastolic	mmHg
Mean arterial pressure	mmHg
Pulse pressure	mmHg

Fig. 4.5 Generated Form in the English Language

The screenshot shows the same web form as Fig. 4.5, but with the language dropdown set to "Japanese". The title "Blood Pressure(OBSERVATION)" and the patient information fields remain the same. The "Any event" section is now titled "任意のイベント" and contains four rows of input fields with Japanese labels: "収縮期" (Systolic), "拡張期" (Diastolic), "平均動脈圧" (Mean arterial pressure), and "脈圧" (Pulse pressure). Each row has a unit dropdown menu set to "mmHg".

Blood Pressure(OBSERVATION)	
Patient ID	4519
Patient Name	Levi
任意のイベント	
収縮期	mmHg
拡張期	mmHg
平均動脈圧	mmHg
脈圧	mmHg

Fig. 4.6 Generated Form in the Japanese Language

Fig. 4.5 shows two attributes of the blood pressure concept in the UI generated in the English language, and Fig 4.6 shows the same two attributes of the blood pressure concept in the UI generated in the Japanese language.

Some other example forms:

- **Health Risk (Type: Evaluation)**

The screenshot shows a web form titled "Health Risk(EVALUATION)" with a language dropdown set to "English". The form contains the following fields:

- Patient ID
- Patient Name
- Health risk
- Risk factors**
 - Risk factor
 - Presence (dropdown menu with options: Present, Indeterminate, Absent)
 - Description
 - Date Ident

Fig. 4.7 Generated Form for Health Risk archetype (English)

Health Risk(EVALUATION) Italian ▾

Patient ID

Patient Name

Rischio sanitario

Fattori di rischio

Fattore di rischio

Presenza ▾

Descrizione

Presente
Indeterminato
Assente

Fig. 4.8 Generated Form for Health Risk archetype (Italian)

- **Health Summary (Type: Composition)**

Health Summary(COMPOSITION) English ▾

Patient ID

Patient Name

context

Start_time 22-04-2023 12:00

Setting

category event ▾

Fig. 4.9 Generated Form for Health Summary archetype (English)

- **Medication Order (Type: Instruction)**

The screenshot shows a web form titled "Medication Order(INSTRUCTION)" with a language dropdown set to "English". The form contains three main sections: "Patient ID" and "Patient Name" at the top, followed by a section titled "Order" which includes three sub-fields: "Medication item", "Route", and "Body site". Each field is represented by a light blue header and a white input box.

Fig. 4.10 Generated Form for Medication order archetype (English)

The screenshot shows the same web form as Figure 4.10, but with the language dropdown set to "German". The form structure is identical, but the labels are in German: "Patient ID", "Patient Name", and a section titled "Verordnung" containing sub-fields "Arzneimittel", "Verabreichungsweg", and "Körperstelle". Each field is represented by a light blue header and a white input box.

Fig. 4.11 Generated Form for Medication order archetype (German)

- **Health Education (Type: Action)**

Health Education(ACTION) English ▾

Patient ID

Patient Name

Ism_transition

Current_state ▾

Transition ▾

Careflow_step ▾

Topic name ▾

suspended

active

aborted

completed

planned

postponed

cancelled

scheduled

Fig. 4.12 Generated Form for Health education archetype (English)

4.3. Persisting form data and Terminology Binding

Blood Pressure(OBSERVATION) English ▾

Patient ID 43526

Patient Name John

Any event

Systolic 120 mmHg ▾

Diastolic 80 mmHg ▾

Fig. 4.13 A sample with a part of Form and some dummy data

```

{
  "archetypeId": "bp-ln",
  "name": "Blood Pressure",
  "rmType": "OBSERVATION",
  "ln": "en",
  "patient": {
    "patient_id": {
      "id": "patient_id",
      "name": "Patient ID",
      "value": "43526"
    },
    "patient_name": {
      "id": "patient_name",
      "name": "Patient Name",
      "value": "John"
    }
  },
  "data": {
    "children": {
      "any_event": {
        "children": {
          "systolic": {
            "termBindings": {
              "SNOMED-CT": {
                "value": "[SNOMED-CT(2003)::271649006]",
                "terminologyId": "SNOMED-CT"
              }
            }
          },
          "inputs": [
            "120",
            "mmHg"
          ]
        }
      }
    }
  }
}

```

Snomed CT
unique ID for
Systolic blood
pressure

Fig. 4.14 Generated JSON response for the form

```

_id: ObjectId('644512201169f0c443472205')
archetypeId: "bp-ln"
name: "Blood Pressure"
rmType: "OBSERVATION"
ln: "en"
✓ patient: Object
  ✓ patient_id: Object
    id: "patient_id"
    name: "Patient ID"
    value: "43526"
  ✓ patient_name: Object
    id: "patient_name"
    name: "Patient Name"
    value: "John"
✓ data: Object
  ✓ children: Object
    ✓ any_event: Object
      ✓ children: Object
        ✓ systolic: Object
          ✓ termBindings: Object
            ✓ SNOMED-CT: Object
              value: "[SNOMED-CT(2003)::271649006]"
              terminologyId: "SNOMED-CT"
          ✓ inputs: Array
            0: "120"
            1: "mmHg"

```

Fig. 4.15 Form data stored in the MongoDB database

Figure 4.13 shows sample data filled by the user, and when the user submits the Form, the system generates a JSON response (shown in figure 4.14), which is sent to the backend server using a POST request and the JSON response is stored (shown in figure 4.15) in the MongoDB database using an insert query.

4.4. Survey of existing projects

Project	Domain/purpose	Interoperability	Dynamic form generation	Language neutrality	Data Persistence
Opereffa [32]	Clinical Application	Yes	No	No	Yes
OpenEHR-Gen Framework [33]	EHR system	Yes	Yes	No	Yes (Using relational Databases)
EHR Flex [34]	EHR system	Yes	Yes	No	Yes (XML based)
EHRServer [35]	Clinical Information management system	Yes	Yes	No	Yes (Using MySQL)
UEHR	Clinical data repository	Yes	Yes	Yes	Yes (Using NoSQL based MongoDB)

Table 4.1 Table of Survey of existing openEHR based projects

In Table 4.1, we present a survey comparing openEHR-based projects and UEHR across multiple criteria, such as Interoperability, Dynamic form generation, Language neutrality, and Data Persistence. We found that none of the existing projects encompassed all these functionalities mentioned in the different columns of Table 4.1. However, UEHR stands out by effectively addressing interoperability, dynamic form generation, language neutrality, and data persistence in a manner unmatched by any other existing project.

CHAPTER 5

CONCLUSION

Challenges	Solution Approach
Syntactic Interoperability	Standardized EHR
Semantic Interoperability	With the help of archetypes and SNOMED CT integration.
Adding support for multiple entries	The UEHR implementation is generic and supports various entries or RM type such as 'OBSERVATION', 'ACTION', 'INSTRUCTION', 'COMPOSITION', 'CLUSTER', etc
Form generation from ADL file	ADL file is used to generate a template, which is later converted to JSON file and this JSON file is fed to a JavaScript code that generates the Form.
Language Neutrality	Language neutrality is provided with the help of translations available in an archetype.
Persisting medical data	To persist the medical data, we have used NoSQL based MongoDB database.

Table 5.1 Summary of challenges and their solution approach.

Table 5.1 provides a comprehensive overview of the challenges we faced throughout the development of the Usability Electronic Health Record (UEHR) system, along with the corresponding solution approaches we adopted to overcome each challenge. This table serves as a valuable reference, showcasing our journey's obstacles and the strategic measures we implemented to ensure the successful development and implementation of the UEHR. It offers insights into the complexities encountered during the development process and demonstrates our commitment to addressing and resolving these challenges in systematically and effectively.

PUBLICATION

The "10th International Conference on Big Data Analytics in Astronomy, Science, and Engineering" [30] has accepted our paper on "UEHR," and we presented our Poster for the conference at the National Institute of Technology, Delhi in collaboration with the University of Aizu, Japan



UEHR – Usability In EHR



Gourav Bansal, Kishan Srivastava, Vardan Agarwal
National Institute of Technology Delhi

Introduction

Electronic Health Records (EHRs) are the digitized version of a patient's health records, providing more accessible and faster access to information to the patient or to a doctor in time of need. With the day-to-day universal acceptance of EHRs, it has become essential to have standardized EHR, enabling the universal exchange of health records more accessible and overcoming the language barrier. Overcoming these challenges require interoperability between the different EHRs and language neutrality. In a country like India, it is essential to adapt the EHRs in regional languages as well, so that the general public can use the technology efficiently.

The standardization of EHRs has helped in achieving the interoperability, with the help of archetypes, which are reusable templates in which medical concepts are represented standardly. Archetypes are written in archetype definition language ("adl") which is very similar to XML.

For a doctor or a patient, archetypes (or templates, which are made by using different archetypes together) doesn't serve the complete purpose as long as they can interact with them easily. An interactive UI as forms can be a solution to this.

Our aim is to generate an interactive form with the help of archetypes, which also supports language neutrality, i.e., it lets user chose between different language then wish to view the form in.

Solution Approach

The following steps are followed in the solution approach:

- Firstly the required archetypes are fetched from the CKM (Clinical Knowledge Manager, which is an open source repository managed by openEHR and it stores the standard archetypes, templates and terminologies) repository.
- In archetypes, language translations are available to almost all the archetype's required sections (like description, details, and ontology).
- To generate a GUI, we convert the archetype into a template using the Archetype Designer Tool.
- That template is passed to AWB Compiler, which converts it into an Archetype Object Model (AOM), and then serialises it into a JSON document.
- This JSON document is passed to our JavaScript program, which extracts all its data attributes, input constraints, and terminologies and uses them to generate a form GUI.
- Input Fields are defined for each data type in the archetype. The data attributes are traversed, and input fields are generated based on the mentioned input constraints.

Results



Figure shows the JSON file created generated from the blood pressure ADL file. The JSON file contains information in two languages- English and Japanese



Figure shows generated form's GUI in English Language



Figure shows generated form's GUI in Japanese Language

Conclusion

This paper aims to solve the problem faced by many medical practitioners and patients due to the unavailability of medical forms in their regional language. The proposed approach provides a way to generate medical forms in many languages.

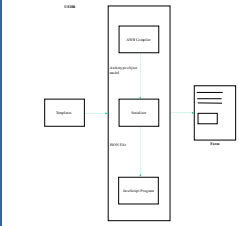
Future Scope

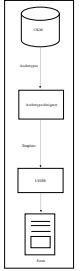
In this paper, we have discussed the approach to generating medical forms in different languages. In our approach, we convert the ADL file of a template into JSON format using the archetype designer. In next iteration, we will work on generating the JSON in our application without the use of archetype designer.

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Proposed Model





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Fig. 6.1 Poster presented at the conference

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